

Questionnaires 101

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Judge a man by his questions rather than by his answers. ~ Voltaire

/// In preparation for this lecture

Find a recent questionnaire, preferably one that applies to you or one that you had to fill out. Think about the questions, what are they asking, what did/would you answer, and how did/would the questionnaire make you feel.

// Summary

There are a limited number of studies on each of these points, we need more, but we need to design questionnaires today and tomorrow and the next day before those studies are done. ~ Krosnick

Questionnaires are the measuring instrument for opinions and play a central role in the data collection process in academic research, market research, policy/political surveys, consumer feedback, employee evaluations, and many others.

The topic of questionnaire design is very broad and would arguably require several courses to cover thoroughly. The goal of this chapter is therefore to:

1. Illustrate the complexity of the topic, so that we do not underestimate it.
2. Cover the most important building blocks, so that we can start viewing others' and our own questionnaires with a critical eye.
3. Provide a starting point and references, so that we may know where to look for details when needed.

We will cover the cognitive process of question answering, introductions and conclusions to questionnaires, types of questions, types of scales, and the pretesting and testing of questionnaires.

Over the course of the chapter we will recognize these overarching themes, which are also key takeaway messages:

- Many decisions have to be made when designing a questionnaire.
- Even minor details in the formulation of questions and answers can have a major effect on the responses obtained and ultimately on the conclusions we draw.
- Research is scarce and most design recommendations are at least a bit controversial – they should be taken as guidelines and not as absolute truths.
- So, when in doubt, the best approach is to test the questions and the questionnaire.

// A cognitive model of the response process

The longer people think, the more confused they get.

In order to develop a better understanding of what is a good and what is a bad question or questionnaire, we must have a good understanding of the participant's experience when answering questions.

For a more systematic treatment we will adopt Tourangeau's four-stage psychological model of survey response. The four stages are comprehension, retrieval, judgement, and response.

Comprehension: First, the respondents must comprehend the question. Many of the most common prescriptions for writing questions are intended to avoid problems with the comprehension stage of the response process by ensuring that respondents can understand what all the words in the question mean and what is being asked of them.

Retrieval: Second, they must retrieve the information needed to formulate a response. Questions should be written to accommodate the retrieval process, which means accounting for how people recall information. People are better able to recall events that happened recently, are distinctive, or that are important. People can also more accurately recall events when the recall period is shorter.

Judgement: Third, they must use that information to make a judgment. These multiple considerations are often weighted and aggregated in complex ways. Some considerations may even be disregarded altogether if they are determined to be irrelevant, invalid, or redundant. Judgement is also impacted through question order effects.

Response: Finally, they must report their answer. Reporting is the final step of the response process in which the respondent actually provides the answer. For the reporting stage to go well, the respondent has to have the ability and the willingness to report her answer, both of which are impacted by a number of questionnaire design features.

// General guidelines

If cognitive demands/load on respondents are minimized and mental capacity is freed up in order to think about a response, the quality of the data obtained is likely to improve.

///// Questionnaire length

Many studies confirm that questionnaire length has a substantial effect on the number of participants that start the questionnaire. Similarly, respondent fatigue is also well documented. Tired or bored respondents more often say that they don't know, choose answers down the same column, give shorter answers, or give up answering the questionnaire altogether.

///// Question length

Keep questions as short as possible. Research for the English language suggests a maximum of 15-20 words. On the other hand, some research shows that it increases data quality if a question or a group of questions is preceded by a medium-length introduction (15-60 words).

///// Grammar and vocabulary

In general, we should keep grammatical complexity at a minimum: active over passive, repeat nouns instead of pronouns (you, he, she, it, we, they), avoid negations.

Use a simple vocabulary. Avoid specialized terminology and complicated words. Don't use jargon, avoid ambiguous words. Always spell out the complete form of abbreviations and acronyms.

///// Be specific

Give all the details concerning the question's frame of reference. A question's frame of reference is one aspect of being specific, but not the only one. In order to get a uniform response from all respondents, the question should be as specific as possible:

Examples and more specific alternatives:

Version 1: What is your income? Version 2: What was your household's total income, from all sources before taxes and deductions, for last year?

Version 1: Do you have any chronic health conditions? Version 2: Do you have any health conditions that required you to see a doctor more than three times in the past 12 months?

Version 1: How would you rank these topics? Version 2: How would you rank these topics from best (1) to worst (5)?

///// Degree of frequency/probability

There is a general consensus that frequently, usually, regularly, etc. (expressions of frequency) and probable, not likely, almost certainly, etc. (expressions of probability) can have quite different meanings for different respondents. We can avoid the problem by using more specific qualifiers, such as at least once per month or between 2 and 5 cigarettes per day.

///// Question order

Similar questions should be grouped together and questions on the same or similar topic should proceed from more general to more specific.

When questions interact their order can have a noticeable effect on the responses.

Examples of effects of question order:

In a study participants were asked (a) whether or not US reporters should be allowed into what was then the Soviet Union, and (b) whether or not reporters from the Soviet Union should be allowed to enter the US. Schumann and Presser (1996) (see Lietz (2010) for details) found that agreement with (b) was significantly greater if (a) preceded (b) than if (b) preceded (a).

Another example where order matters are demographics questions. It is common practice to move demographic questions to the end, to avoid the provision of personal information to have an effect on the responses to further questions. This logic can be applied more generally, to other sensitive questions.

Similarly, early questions should be easier and pleasant to answer to build a rapport with the respondent.

A **filter question** (or screening question) is a question that is used to avoid asking respondents further questions that do not pertain to them. Filter questions should always be included to avoid asking unnecessary questions.

// Front and backmatter

//// Introduction

The introduction is the respondent's first contact with the questionnaire is thus very important. While this might vary from study to study, the introduction should:

- provide the subject of the survey,
- identify the surveyor,

- explain the purpose of the survey,
- request the respondent's co-operation,
- explain how much time it will take,
- establish confidence,
- inform the respondent about confidentiality issues, the status of the survey (voluntary or mandatory), and any existing data-sharing agreements with other organizations.

Respondents frequently question the value of the gathered information to themselves and to others. Therefore, be sure to explain why it is important to complete the questionnaire, how the information will be used, and how respondents can access the results. Ensuring that respondents understand the value of their information is vital in undertaking a survey.

Exercise:

Below is an example of an introduction from the 2021 internal survey at UL-FRI. What does the introduction do well? Could it be improved and how?

Satisfaction with faculty services – students

Dear students!

We kindly ask you to participate in a survey on satisfaction with the faculty services which you used during your studies (student affairs office, library and international office) in 2020/2021 academic year.

Important: If you are dissatisfied with a specific service, we kindly ask you for a constructive comment that will help the service to improve. A low score alone does not say much about the cause of dissatisfaction.

The survey is anonymous and will be active until Nov 7 2021.

Thank you in advance for all the helpful information you will contribute.

Sincerely,

(signed by the dean of the faculty)

//// Opening questions

The opening questions of any survey should establish the respondents' confidence in their ability to answer the remaining questions. If necessary, the opening questions should help determine whether the respondent is a member of the survey population. For example, if the survey is for students only, that should probably be the first question.

//// Conclusion

A good questionnaire ends with a comments section that allows the respondent to record any other issues not covered by the questionnaire. This is one way of avoiding any frustration on the part of the respondent, as well as allowing them to express any thoughts, questions, or concerns they might have. Lastly, there should be a message at the end thanking the respondents for their time and patience in completing the questionnaire. Contact information can again be provided even though we already gave it in the introduction.

// Questions

//// Open and closed-type questions

There are two main types of questions: **open** and **closed**. Open questions give respondents an opportunity to answer the question in their own words. Closed questions give respondents a choice of answers and the respondent is supposed to select one.

Open questions:

- Allow the respondent to interpret the question and answer how they choose.
- Their use has been declining over time.
- Common criticism (without empirical support): results depend on articulation ability of respondent, susceptible to salience effects.

Salience = the quality of being particularly noticeable or important. **Salience bias** = our tendency to focus on items or information that are more noteworthy while ignoring those that do not grab our attention. However, literature shows it's not that different for open or closed questions.

Example:

Suppose we ask the respondent what they think the most important problem facing the country is. If they happened to have seen a news story about crime on television last night maybe that enhances the likelihood that you would retrieve crime as a potential problem to answer with whereas, with a closed question on a list, maybe that salience effect will be minimized.

Closed questions:

- Provide some consistency across respondents.
- Easier and faster for the respondent to answer. Typically twice as much time.
- Easier and faster for the researcher to code and analyze.
- Problems with non-attitudes: Non-attitude refers to the mental state of having no attitude or opinion toward some object, concept, or other type of stimulus. In survey research, this is manifested by an overt no opinion or don't know response to an attitude question, but it may also be hidden by a random or guesswork choice of answers to avoid appearing ignorant. Additionally, it is likely that not all no opinion or don't know responses reflect non-attitudes. This makes it hard to estimate how many respondents have non-attitudes toward the object.
- Gravitation towards positive.
- Incomplete universe problem: "Other" does not solve the incomplete universe problem, offering that other option does almost nothing (people almost never select it).

In practice, open-ended questions should be asked when we are not sure about the universe of possible answers to a categorical questions. Alternatively, we can pre-test to obtain a universe.

/// Notable groups of questions

///// Sensitive questions

Sensitive questions is an umbrella term for questions where there is a substantially higher probability that respondents will be untruthful with their response. Typically, these are questions that could make people feel ashamed, stupid, ignorant, afraid, or some other negative emotion. It is also possible to get untruthful responses due to invoking positive emotions.

Sensitive questions can be as simple as establishing questions (age, race, income), they can be an invasion of privacy (politics, sexual activities), they may put the respondent in a position where they might reveal illegal or socially undesirable behavior and thus pose a risk (drug use, cheating on

tests, tax fraud), or they can put the respondent in an emotionally stressful situation (grief, disability). Note that people are more inclined to provide socially desirable answers, so questions that endorse unpopular behaviors or attitudes (anti-vaccine, racism) or socially desirable (mask wearing, inclusion) are more likely to be answered untruthfully.

Examples of sensitive questions:

How often do you exercise?

What is your salary?

What was your exam score?

Did you vote in the last election?

Do you take drugs?

Are you vaccinated against COVID-19?

There are several ways of dealing with sensitive questions:

- Provide an “out” answer, such as I prefer not to answer.
- Anonymity or at least confidentiality of the study might increase response rates. Of course, respondents must be made aware of this in the frontmatter.
- Indirect questioning: What do you believe other people think about insert sensitive topic?
- Move the sensitive questions towards the end: This serves two purposes. First, you lead up to sensitive questions with less sensitive questions thus establishing a rapport with the respondent. Second, when respondents have already invested some time in the survey they are often more likely to complete the survey.
- Use of ranges instead of specific numbers, where appropriate (age, salary).
- Use of question loading to normalize behavior. That is, prefix the question with text that removes at least some of the judgement associated with the negative or sensitive response.

Examples of question loading for normalization:

There are many reasons why people don't get vaccinated, such as medical considerations, religious beliefs, and ethical concerns. Are you vaccinated against COVID-19?

Recent studies have shown that people often have too much work to exercise regularly. How often do you exercise?

Most UL-FRI students fail at least one exam and it is not uncommon for a student to perform poorly on an exam even though they were well prepared and are knowledgeable. What was your exam score?

Related to sensitive questions is the use of emotionally charged words that can stir up unnecessary emotion in the participant. For example, words such as elite, patriotic, horrible, crisis, joy, fear, sadness, etc. The use of such words is appropriate in literary works, advertising campaign, etc., but should be avoided in questionnaires. A special kind of emotionally charged words are prestige names – names of people, companies, locations, etc.

Examples of emotionally charged words:

Do you support Trump's new foreign policy?

How much do people that refuse to get vaccinated contribute to this horrible epidemic?

How happy were you with this amazing result?

///// Double-barreled questions

A double-barreled question is when someone asks a question that touches upon more than one issue, yet allows only for one answer.

Examples of emotionally charged words:

Do you like cake, chocolate and other sweets?

How satisfied are you with your work environment and compensation?

Please rate the instructors preparation/materials.

///// Leading questions

We have already discussed leading questions in the context of sensitive questions, where they can be used to normalize potentially sensitive behavior in order to increase the probability of a truthful response. In a similar way, questions can be designed to lead the respondent not towards a truthful response but instead towards a particular response. We may do this intentionally or unknowingly, but either way it introduces a bias and should be avoided.

Example taken from the TV show Yes, Prime Minister S01E02, "The Ministerial Broadcast"

Sequence 1:

Are you worried about the number of young people without jobs? – Yes

Are you worried about the rise in crime among teenagers? – Yes

Do you think there's a lack of discipline in our society? – Yes

Do you think young people could use more authority and leadership in their lives? – Yes

Would you be in favor of reintroducing the national service? – ???

Now consider Sequence 2:

Are you worried about the danger of war? – Yes

Are you worried about the growth of armaments? – Yes

Do you think there's a danger in giving young people guns and teaching them how to kill? – Yes

Do you think it's wrong to force people to take up arms against their will? – Yes

Would you oppose the reintroduction of national service? – ???

///// Loaded questions

A loaded question is a question that contains a controversial assumption, something that the participant might not agree with or that might not even be true. By answering such a question, the

participant would put himself into an unfavorable position. We should avoid such questions.

Examples of loaded questions:

Why do you think that the Bayesian statistics course is the best course?

What's your favorite part about working with our support team?

Should a smack as part of good parental correction be a criminal offence in New Zealand?"

Do you get aroused when you put on women's clothes?

What other benefits do you see for having a dog, other than for food?

//// Negatively worded questions

A negative worded question is defined as a question in which disagreement would be a positive answer. Evidence shows that negatively worded questions take longer to process, can lead to more negative responses, and can confuse the respondents into making mistakes.

Examples of negatively worded questions:

Should stores not be allowed to be open on Sundays?

Do you think the government should not be so uncooperative?

// Responses and response scales

//// Scale length

In practice, scales can vary from 2 to even 100 points. In general, more items lead to more expressiveness but potentially more confusion, because at some point it becomes more difficult to discern, for example, what is the difference between 65 and 67 on a 100 point scale.

Shorter scales have less information and responses have to be treated as ordinal or even as categorical. As the scales lengthen, we are more and more able to analyze them as numerical, even if they are discrete. Research shows that there might be very little gain from as few as 4 points onwards. A general consensus is that 6 should be enough, because information that pertains to internal consistency doesn't grow anymore.

For respondents with more relative cognitive ability, more points may lead to more information. For the general public, 5 points should be enough in most cases.

Bipolar (like-dislike) and unipolar (not important – extremely important) scales seem to differ in the optimal number of points (7 for bipolar, 5 for unipolar).

In most cases a longer scale leads to longer response times. There is one notable exception: counter-intuitively, 5 point scales have lower response times than 4 and 6 point scales.

//// Scale labels

We want to avoid inflation of responses on the upper or lower bound of the scale. If we anticipate a skewed distribution, we should transform the scale. Note that respondents have a tendency to avoid negative answers when grading things, in particular people.

There isn't a lot of concrete evidence that mid-points (odd number scales) are bad. It turns out that offering mid-points is a good thing, the majority of people who select them do so because they're indeed neutral and they are not doing so to satisfice. To satisfice is to decide on and pursue a course of action that will satisfy the minimum requirements necessary to achieve a particular goal.

//// Branching

Empirical evidence shows that branching a bipolar scale is better than a single long scale. If there is a precise midpoint, we should not branch the midpoint, instead, we should include the midpoint in the first question. If there is no precise midpoint, the first question has only two possible responses.

Examples of branching questions:

Example 1:

Q1a: Should the amount of homework in this course increase, decrease, or remain the same?

Q1b: How much should it increase, a little, moderately, or a lot? (assuming answer to Q1a was increase)

Example 2:

Q2a: Was your experience in the hotel positive or negative?

Q2b: Was it a little positive, positive, or very positive? (assuming answer to Q2a was positive)

//// Other interesting and counter-intuitive results

- People are more likely to answer positive on a bipolar scale than they are to answer more than 5 on a unipolar 0-10 scale.
- Direction of response: It doesn't seem to matter if the scale is left to right (agree to disagree) right to left (disagree to agree).

// Questionnaire pretesting/testing

Testing is a fundamental step in developing a questionnaire. It helps us, among other things, to discover poor wording or ordering of questions, identify errors in the layout and instructions, determine the respondent's inability or unwillingness to answer the questions, suggest additional response categories that can be pre-coded on the questionnaire, and provide a preliminary indication of the time necessary to complete the questionnaire.

Testing can include the complete questionnaire or only a particular part of it. Ideally, the complete questionnaire will at some point also be tested. Because there are so few absolute truths about designing a questionnaire, testing is even more important.

The two key terms that we will be operating with are **reliability** (are the measurements reliable, consistent; variability) and **validity** (is it measuring what it claims to measuring; bias).

Examples to highlight the difference between reliability and validity:

A textbook example is a clock that doesn't run. It is absolutely reliable, but not at all a valid measurement of time. On the other hand, a clock that tends to be a few minutes early or late is not very reliable, but is a valid measurement of time.

Measuring tape is a relatively reliable (plus-minus a few millimeters) device and it is a valid

measurement of length. Measuring tape is also a valid tool for measuring the circumference of a person's skull. But, contrary to what a lot of people thought 150 years ago, the circumference of the skull is not a valid measurement of intelligence.

University professors' grades are by definition a valid measurement of the student's performance in a particular course. However, they are in most cases very unreliable, because choice of exam questions, daily form, professors' error and bias, etc. can have a strong effect. Furthermore, it can be argued that grades are not a valid measurement of the student's academic ability.

//// Testing reliability

Testing reliability is synonymous with identifying sources of variability and quantifying them. The two most commonly investigated aspect of reliability is how much the measurement varies if applied to the same participant (test-retest or intra-rater reliability).

If a question is an unreliable measurement of a quantity or construct (for example, personality), we typically include more questions that measure the same quantity or construct. The most common validation is to show the internal consistency of such a questionnaire.

When multiple participants represent repeated measurements of the same objects or subjects, such as rating an item on Amazon, providing feedback on coworkers, or grading student projects, we can and should also investigate inter-rater reliability. If inter-rater reliability is low (if inter-rater variability is high), there is typically no other alternative than to increase the number of raters. Similar to intra-rater variability, modeling inter-rater variability is a problem of estimating the distance between raters' responses.

In most cases, analyzing inter-rater/intra-rater reliability and internal consistency are relatively simple statistical modeling tasks, assuming that the data were gathered correctly. Examples are given at the end of this chapter.

//// Testing validity

Even the most reliable measuring tool is useless if it doesn't measure what we are interested in. Testing the validity of a tool is relatively simple when a gold standard or ground truth reference is available. However, when we are trying to measure a latent (unobserved) variable and no reference is available, we can only justify its validity indirectly. Fields such as psychology, sociology, etc. predominately face problems like this. From these fields comes the term **construct**, which is basically a conceptualization of a latent variable (intelligence, empathy, power, likeability). We can't touch intelligence, but we agree that it exists, because we observe consequences of it. That is, measurable things behave as if there is something like intelligence affecting them.

The overarching term in testing validity is **construct validity**. Construct validity is the approximate truth of the conclusion that your operationalization accurately reflects its construct. Evidence of construct validity must be accumulated over time. There are several aspects of construct validity, but we will name just three of the most general ones:

- **Face validity:** Respondents think that the questions measure what we want to measure. That is, the questionnaire makes sense at face value.
- **Content validity:** To simplify, experts agree that the questionnaire measures all aspects of the construct. Both face and content validity can be tested only qualitatively.
- **Criterion validity:** The construct measurements should correlate with survey questions or external data about the respondents (concurrent validity) or with data gathered in the future (predictive validity). For example, two of the most commonly used constructs, intelligence and personality type, have been thoroughly validated by showing that the measurements correlate with and are able to predict future behavior, business success, graduation rates, etc.

//// Types of pretesting

Stand-alone pretesting: Predominately qualitative approaches:

1. Cognitive interviewing (verbal probing, thinking aloud; typically very expensive) and
2. usability testing (for more interactive questionnaires; such as computerized questionnaires).

Embedded pretesting methods: In contrast with the stand-alone pretesting approaches of cognitive and usability testing, researchers may rely on procedures that are embedded within a field pretest:

1. Psychometric tests (test-retest; internal consistency),
2. planned experiments,
3. field-based probing/debriefing.

// Important topics not covered

Visual design: Good question wording can do a great deal to help respondents understand and answer questions, but it may not be enough. Questions also need to be visually designed to support question answering. Considerable research on visual design of questions and questionnaires has been conducted.

Translating questionnaires: The two main sets of issues with translating questionnaires are the issues with the translation itself, which is not unique. The main techniques are to use multiple translators and to use backtranslation (translating and then translating back) to gauge the appropriateness of the translation. The second main issue is that psychological constructs and questionnaire design choices regarding scales, etc. don't necessarily transfer to other languages and cultures.

We will cover some other survey methodology topics that are closely related to questionnaires, such as survey mode (online, in-person, etc.) and non-response bias in the Sampling chapter.

// Tools for questionnaires

- [1ka](#)
- [SurveyMonkey](#)

// Readings

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